
Categories of Parallel Sorting Algorithms

Comparison-Based Algorithms:

- Operate similarly to traditional sort algorithms.
- Examples:
 - Parallel Merge Sort
 - Parallel QuickSort
 - Bitonic Sort
 - Sample Sort

Non-Comparison-Based Algorithms:

- Use data characteristics (e.g., value ranges, bit representations).
 - Examples:
 - Parallel Radix Sort
 - Counting Sort (for integer data)
 - Bucket Sort
-

When to Use Which Algorithm?

Small Datasets / Few Processors:
Parallel Merge Sort or QuickSort

Sorted Input / Fixed Patterns: Bitonic Sort

Numeric Input / GPU Systems: Radix Sort

Massive Datasets in Clusters: Sample Sort with MPI

Parallel Merge Sort



Divide array into equal partitions for each processor.



Each processor sorts its subarray independently (in parallel).



Parallel merging occurs in $\log_2(P)$ stages.



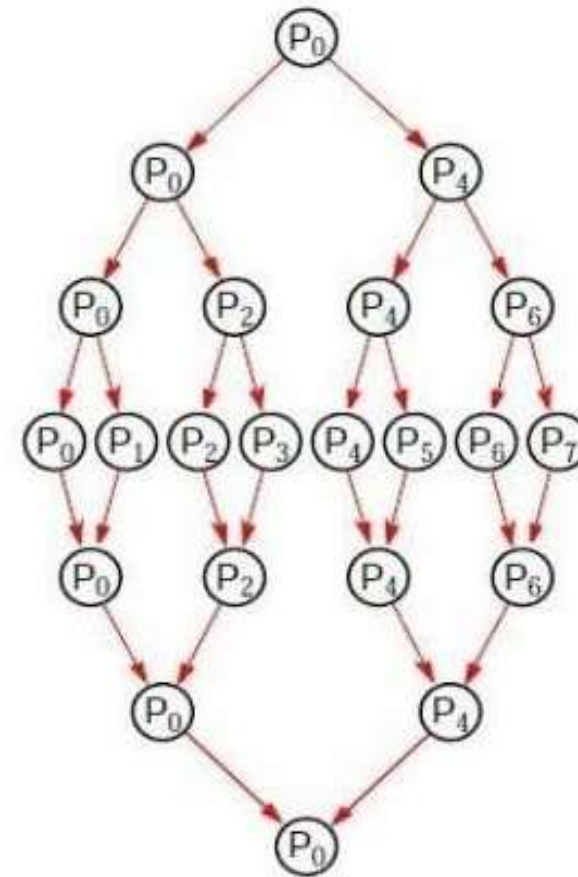
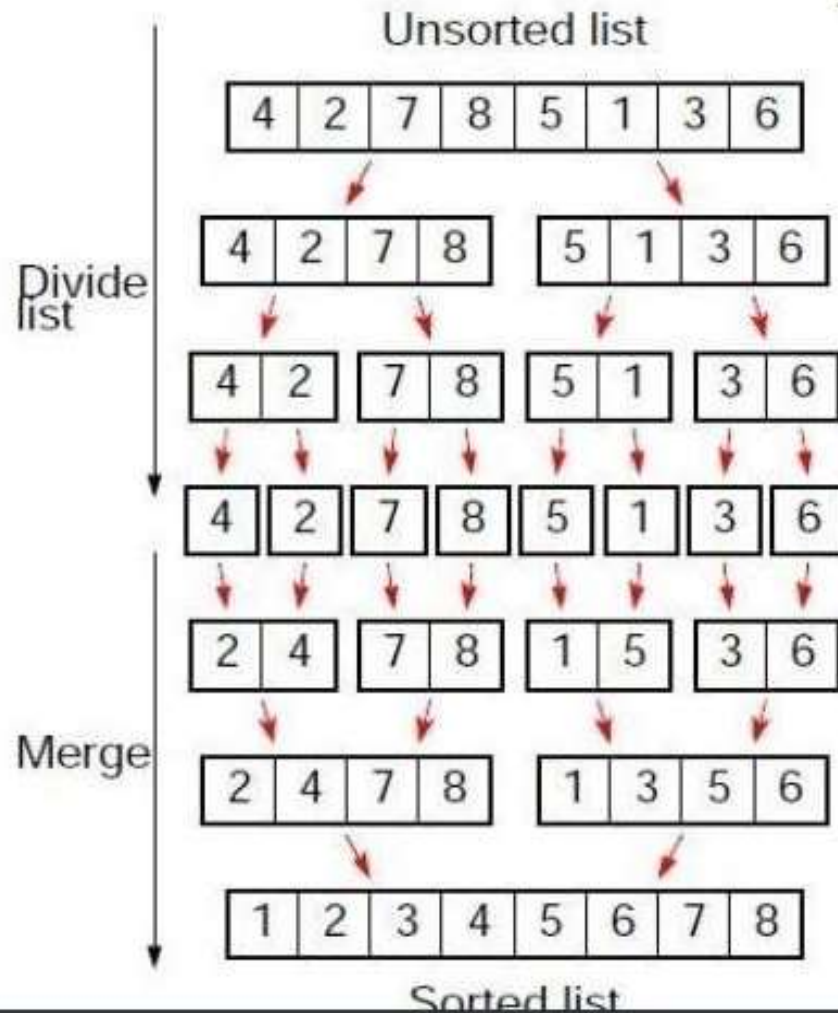
Time Complexity: $O(n/P * \log(n/P) + \log P)$



Efficient for shared-memory systems; merge phase is key bottleneck.

Parallel Merge sort:

Using a strategy to assign work to processors organized in a tree.



Process allocation

Parallel QuickSort

Parallel pivot selection and partitioning.

Each partition is recursively sorted in parallel.

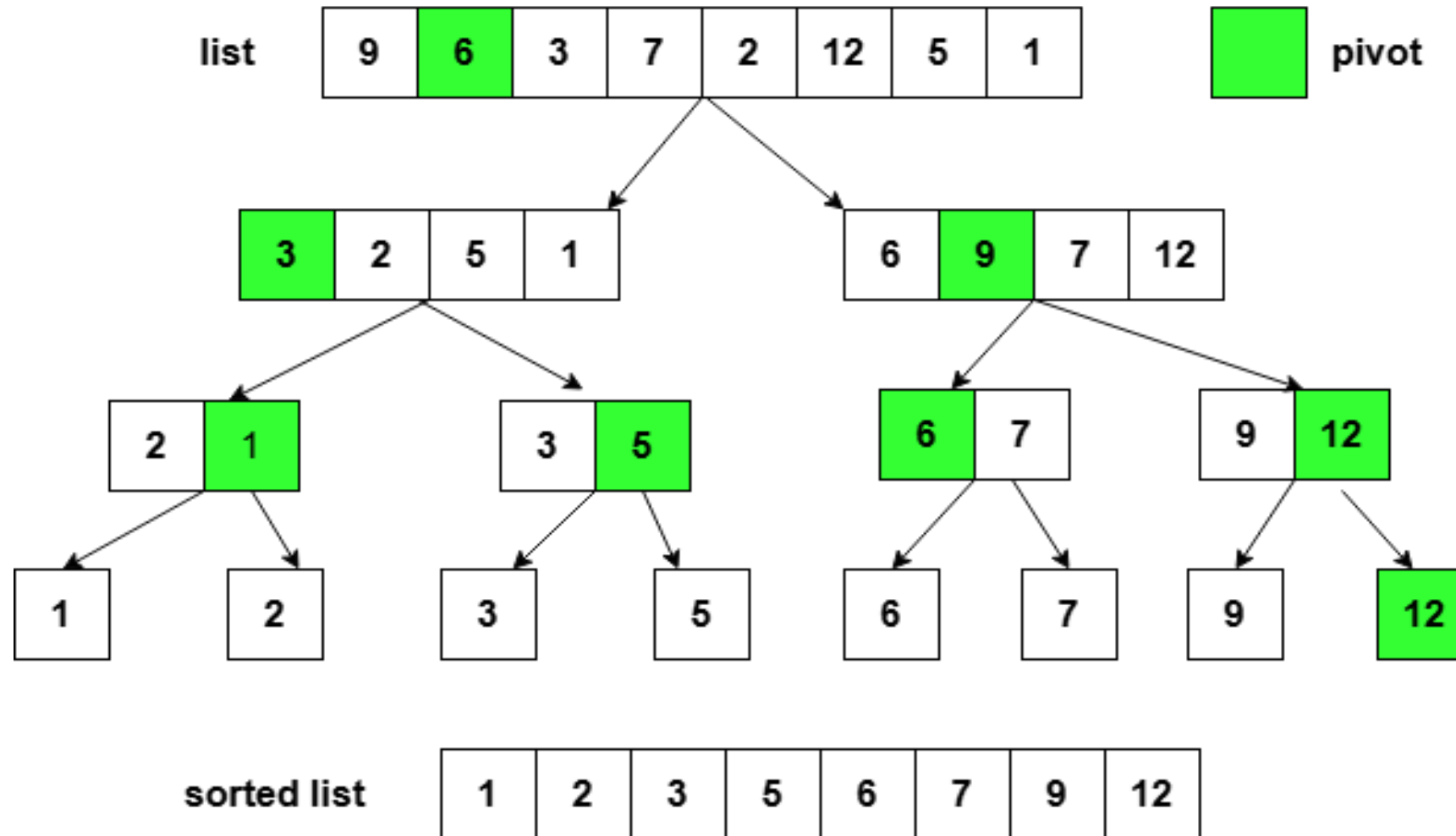
Highly efficient with balanced partitioning.

Time Complexity: $O(n \log n / P)$

In-place sorting with minimal memory overhead.

Solved Example 1

Sequential Quick Sort



Solved Example 2

